Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)		The [minimum] energy required to liberate an electron [from the metal's surface]	1			1		
	(b)		Zinc as {more energy / higher frequency / lower wavelength} required		1		1		
	(c)	(i)	$E_{\mathrm{k\,max}} = hf - \phi$ **and** use of $f = \frac{c}{\lambda}$ (1) @ threshold $E_{\mathrm{k\,max}} = 0$ **and** convincing algebra (1)	2			2	1	
		(ii)	$\lambda_{\mathrm{max}} = \frac{hc}{3.7\times10^{-19}} = 5.4 \times 10^{-7}$ [m] i.e. 540 n[m] (1) 650 nm (accept red) **and** 550 nm (accept green) (1) **Alternative:** Energy of green photon = $\frac{hc}{\lambda_g} = 3.6 \times 10^{-19}$J **AND** energy of blue photon = $\frac{hc}{\lambda_b} = 4.4 \times 10^{-19}$J [Energy of red photon = $\frac{hc}{\lambda_r}$ $= 3.1 \times 10^{-19}$J] (1) 650 nm (accept red) **and** 550 nm (accept green) (1)		2		2	1	
		(iii)	Greater intensity is more photons [s^{-1}] (1) Intensity has no effect on photon energy (1) so max kinetic energy of electrons doesn't change **Conclusion must be present to award both marks**		2		2		
			Question 6 total	**3**	**5**	**0**	**8**	**2**	**0**